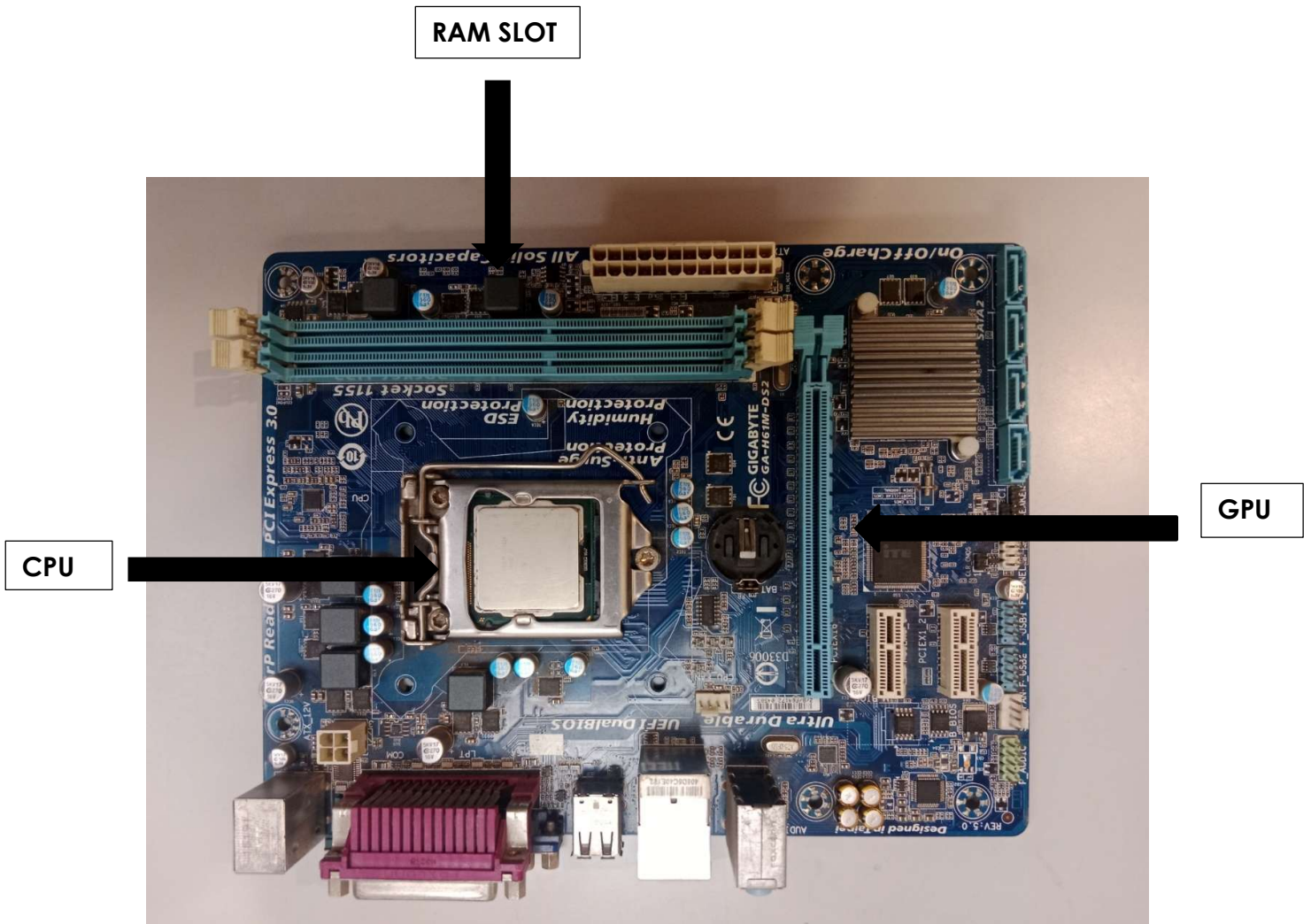
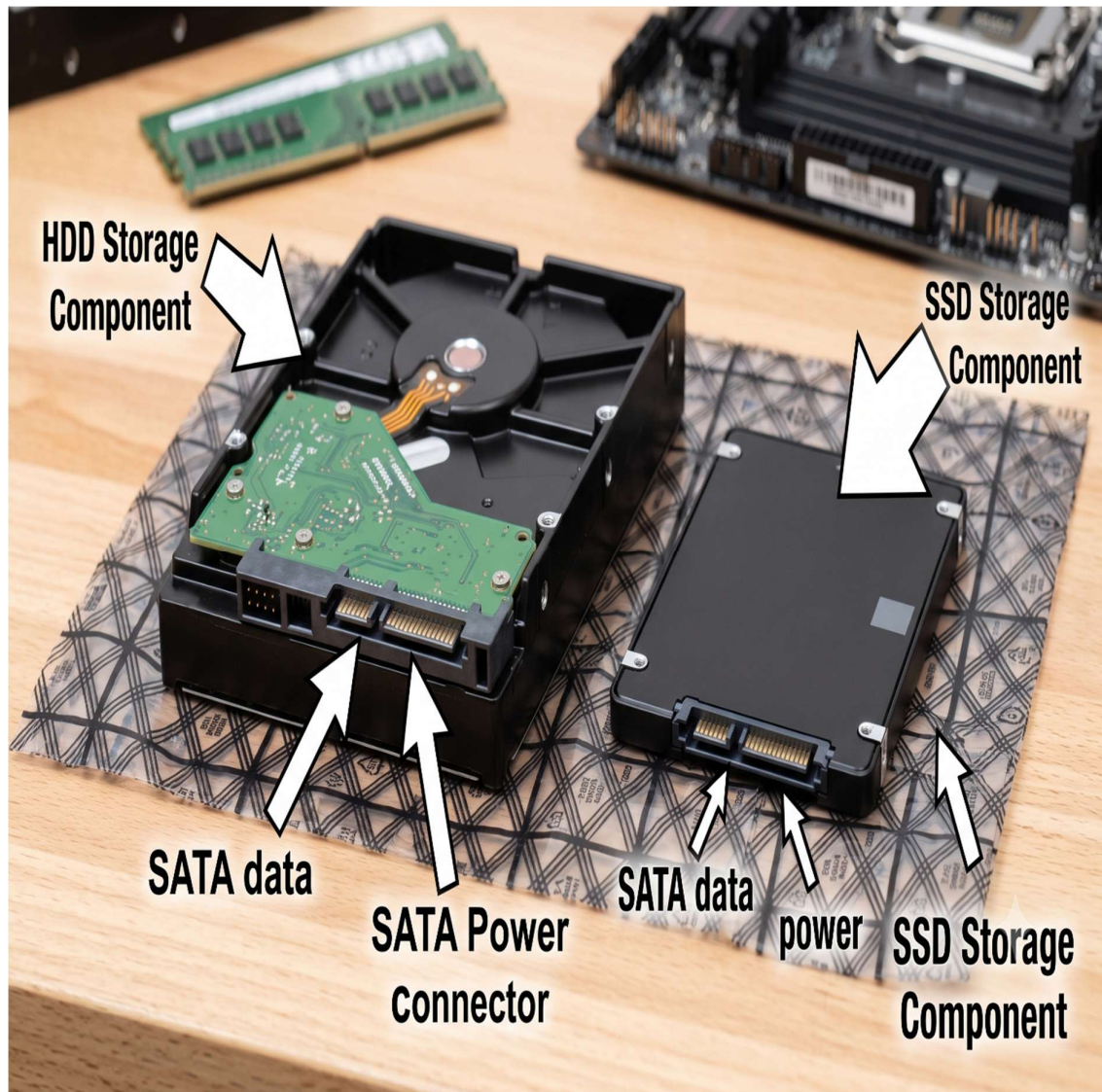


Task 1:

Labeled Diagram of a PC



HDD AND SSD



Task 2:

Comparison Table (HDD vs SSD Storage)

Feature / Criteria	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slow: It uses mechanical spinning platters and a moving read/write head, which takes more time to load files.	Very Fast: It uses flash memory chips (like a pendrive), which allows files and the operating system to load instantly.
Durability	Low: It is fragile because of its moving mechanical parts. If it experiences a shock or is dropped, it can easily get damaged or lose data.	<u>High: It is highly durable and shock-resistant because it has no moving parts, making data completely safe from physical vibrations.</u>
Typical Use Cases	Commonly found in Budget Laptops, old desktop PCs, and external backup drives for cheap, bulk data storage.	<u>Commonly found in Gaming PCs, premium laptops, and used as a Boot Drive (C: Drive) to speed up Windows and software performance.</u>

Task 3:

Role of CPU, RAM, and GPU in Gaming (PUBG / Free Fire)

- CPU (Processor): It handles all the core game logic, physics calculations, and real-time player positioning. It processes the background operations to ensure the game mechanics run without freezing.
- RAM (Memory): It temporarily stores active game files, maps, and textures for quick access by the system. Having sufficient RAM allows game environments to load fast and prevents mid-game stuttering or drops.
- GPU (Graphics Card): It is responsible for rendering all the 3D graphics, visual effects, and environmental lighting on the screen. It directly determines the smoothness of the gameplay by pushing higher Frames Per Second (FPS).

Task 4:

PC Upgrade Plan for Running FIFA Smoothly

- **1. GPU (Dedicated Graphics Card):** FIFA requires heavy 3D rendering for realistic player models, stadium crowds, and smooth animations. Upgrading to a dedicated GPU (like an NVIDIA GTX/RTX series) ensures the game runs at stable 60+ FPS without graphical stutters.
- **2. RAM (System Memory):** Modern games demand a lot of temporary memory to handle background physics and fast transitions. Upgrading from 8GB to **16GB DDR4/DDR5 RAM** will prevent random mid-game lags and frame drops.
- **3. Storage (SSD - Solid State Drive):** FIFA has massive file sizes and long loading screens. Upgrading from an old HDD to a fast **NVMe M.2 SSD** will drastically reduce game boot times, match loading times, and improve overall system responsiveness.

